PART II: PATTERNS

Topic 12: Short-Term Fluctuations vs. Long-Term Change

The Core Concept

Short-term fluctuations are the month-to-month "zigs and zags" caused by weather, holidays, random chance, or local events. **Long-term change** is a structural shift in the data that persists across months and years.

Why It Matters

Reacting to short-term fluctuations as if they were long-term change is one of the most common errors in public safety data analysis. It leads to policy overreaction, wasted resources, and eroded trust in data-driven decision-making.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

Short-term view: Every summer, Precinct North's numbers go up. Every winter, they go down. A manager who only sees this year's data might think a summer crisis is a new problem.

Long-term view: The peaks are declining year over year (42 → 35 → 25) and so are the valleys (25 → 18 → 10). This is long-term structural improvement.

The same fluctuation looks like a "crisis" or "success" depending on whether you look at one month, one year, or three years. Context determines meaning.